

Qatar-4b

R-2714

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_250910 · created 2026-09-01T01:25:22+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460928.9447 +/- 0.0051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.044
Transit depth (Rp/Rs)^2	2.54 +/- 1.39 [%]
Orbital Inclination (inc)	85.35 +/- 1.94
Airmass coefficient 1 (a1)	1.139 +/- 0.038
Airmass coefficient 2 (a2)	-0.101 +/- 0.038
Scatter in the residuals of the lightcurve fit is	3.35 %
Best Comparison Star	#3 - [70, 420]
Optimal Method	PSF photometry
Transit Duration (day)	0.078 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.159 ± 0.044 vs 0.138 (deviation 0.47σ)
Duration fit vs published	0.078 d vs 0.0892 d (deviation 0.69σ)
Mid-time O–C	-13.0 min
Depth SNR	3.6
Residual scatter	3.35 %

Light curve

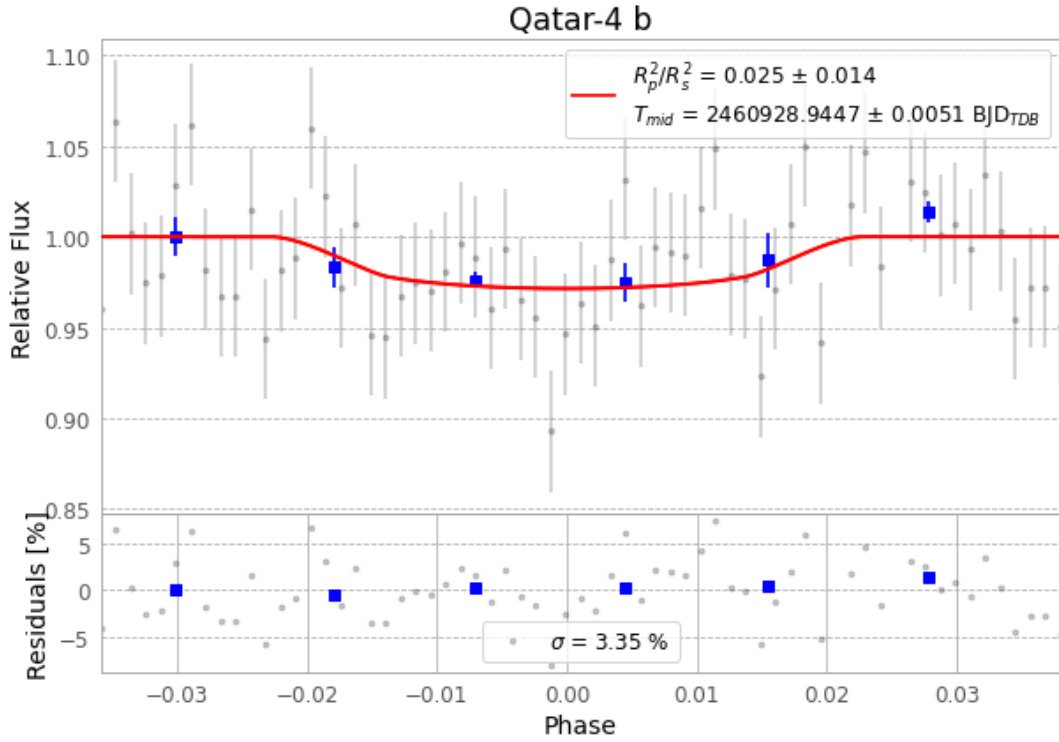


Figure 1 — FinalLightCurve_Qatar-4 b_2025-09-10.png (SHA-256 3b58a3566f8262ac...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [324, 261], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 33bdac2fa46d35ee...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), Qatar-4250910090016.FITS → Qatar-4250910121216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-250910.log (6f3f4dd9cfb3...), FinalLightCurve_Qatar-4 b_2025-09-10.png (3b58a3566f82...), FinalLightCurve_Qatar-4 b_2025-09-10.csv (7cd603bcbb33...)

Compute resources

Wall time 113.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 01:25Z.